Sahaj Bhadja

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Summary

Full-stack engineer with 2+ years building **Python/FastAPI, React/TypeScript, PostgreSQL, and AWS applications**; experienced in production deployments, containers, Linux/Nginx, database performance, scheduled integrations, LLM reliability, CI/CD, observability, and documentation.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **FastAPI and PostgreSQL production services** with normalized schemas, incremental data workflows, indexing, and query diagnostics, supporting 500+ users while reducing dashboard latency from 1.2 seconds to 350 milliseconds.
- Maintained scheduled integration jobs processing 5,000+ recurring monthly workflows with validation, reconciliation, explicit states, and failure recovery, sustaining 99%+ reliability while improving asynchronous processing throughput 3x.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Shipped **responsive React interfaces and Python/Flask APIs** on AWS for uploads, dashboards, processing, and reporting, serving 150+ users and 10,000+ daily requests with stable production performance.
- Managed OpenAI, Claude, and Qwen integrations with reliable token-cost tracking, configurable providers, and fallback handling, achieving 90%+ grading consistency while reducing feedback turnaround 60% through automated workflows.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed **scalable scheduled API and data synchronization jobs** across Python, AWS Glue, S3, SQL, and microservices, improving response times 30%, pipeline efficiency 68%, and storage costs 15%.
- Maintained Linux-compatible Docker builds, integrations, and 20+ unit and integration tests through GitHub and Jenkins CI/CD, generating structured failure logs, coverage reports, and technical documentation for stakeholders.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **production React, TypeScript, and Node.js operational tools** across eight services for records, resources, billing, permissions, notifications, and reporting, securely synchronizing 2M+ records through APIs and WebSockets.
- Resolved internal and customer issues using Sentry, Azure Monitor, Kibana, automated tests, and production logs, achieving 95%+ build validation while reducing deployment time 50% and resource waste 42%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Developed **secure React/TypeScript interfaces and Flask integrations** for booking, pricing, payments, and reporting workflows, supporting 50,000+ records through Firebase synchronization, Databricks analytics, and sub-200-millisecond API responses.
- Worked directly with customers to resolve issues and maintain 12+ external integrations, documenting architecture and workflows while improving synchronization speed 40% and reducing manual refund handling 70%.

Skills

Application: Python, FastAPI, Flask, Django, React.js, TypeScript, Node.js, REST APIs, component libraries
Database: PostgreSQL, MySQL, schema design, data modeling, indexing, query tuning, connection diagnostics
AWS & Systems: EC2, RDS, S3, Docker, Linux, Nginx, scheduled jobs, external integrations
Delivery: GitHub Actions, Git, Jenkins, CI/CD, Terraform, testing, deployments, technical documentation
Reliability: LLM APIs, cost tracking, fallback handling, Sentry, metrics, structured logging, production support

Projects

Big-Brain Live | *PostgreSQL, APIs, Cloud Storage*

- Built a **production data and document platform** processing 50,000+ records with authenticated uploads, PostgreSQL metadata, incremental indexing, validation, REST APIs, cloud storage, and performance tuning, reducing retrieval latency 40%.

Chat2SQL Code | *PostgreSQL, Docker, Integration*

- Enhanced a **cross-platform data-management application** supporting PostgreSQL and 10+ systems with schema exploration, structure synchronization, query generation, compatibility checks, Docker configurations, and reusable integration, deployment, and debugging workflows.

HMIS Live | *Healthcare Tools, TypeScript, Monitoring*

- Developed a **healthcare operations application** with role-based access, dashboards, appointments, REST integrations, notifications, and Sentry monitoring, halving record retrieval time while supporting 100+ daily workflows and reducing administrative work 30%.